DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

1 Why DBSCAN?

- Many clustering methods (like K-Means) assume clusters are spherical and need you to give the number of clusters.
- **DBSCAN** solves this by:
    1. Discovering clusters of arbitrary shapes.
    2. Automatically separating noise/outliers.
    3. Needing only two parameters instead of a fixed number of clusters.

2 Core Concepts

Term	Meaning
eps (ε)	Radius of the neighborhood around a point.
minPts	Minimum number of points required inside eps for a region to be considered dense.
Core point	A point with at least minPts neighbors (including itself) inside distance eps.
Border point	Lies within eps of a core point but doesn't have enough neighbors to be core itself.
Noise / Outlier	Not a core or border point.

3 How DBSCAN Works (Algorithm)

- **Pick an unvisited point**.
- Find all points within distance eps → its neighborhood.
- If neighborhood size ≥ minPts → mark as **core** and create a new cluster.
    1. Expand the cluster by recursively including:
        - All neighbors of the core point.
        - Their neighbors, if they're also core.
- If neighborhood size < minPts → temporarily label as noise.
    2. (May later become border if found inside another core's neighborhood.)
- Repeat until all points are visited.

4 Choosing Parameters

- **eps**: Often decided by a "k-distance plot." Look for the "elbow" where distance sharply rises.
- **minPts**:
    1. Rule of thumb: minPts = dimensions + 1 (or higher for noisy data).

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5 Strengths

- Finds clusters of **any shape** (not just circles).
- Handles **noise/outliers** naturally.
- Doesn't require you to predefine number of clusters.

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6 Limitations

- Sensitive to choice of eps and minPts.
- Struggles when clusters have very different densities.

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7 Python Implementation (High-Level)

```
from sklearn.cluster import DBSCAN

model = DBSCAN(eps=0.5, min_samples=5)
model.fit(X)

labels = model.labels_  # -1 indicates noise
```

- Visualize results to see clusters and outliers.

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8 Intuition Recap

Think of DBSCAN as:

- **Pouring water** on data points:
    1. Dense "puddles" become clusters.
    2. Sparse isolated drops remain as noise.
